



FlashGuard: GRU Based Real-Time Flash Crash Detector

Predicting market microstructure anomalies in Indian equity markets using minute-level OHLCV data

What are Flash Crashes?

Sudden Extreme Drops

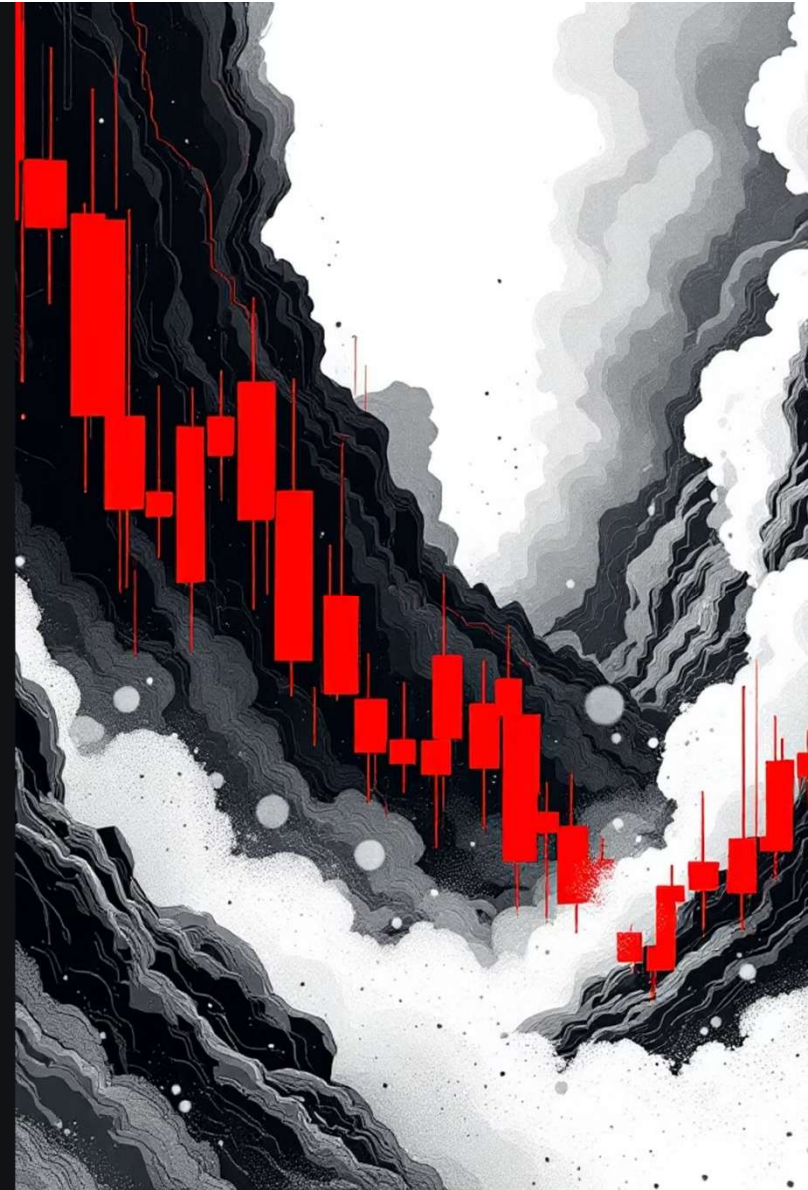
5–15% price drops occurring within minutes, followed by rapid recovery

Algorithmic Causes

Caused by algorithmic trading, HFT feedback loops, and market fragility

Real Examples

June 4, 2024: NIFTY 50 dropped ~1,400 points in 30 minutes





Why This Matters

The Impact

- ₹5+ lakh crore wiped from Indian markets in June 2024 crash
- Individual investors suffer panic-driven losses
- Current surveillance systems are reactive, not predictive

The Gap

No real-time early warning system exists for retail investors

Project Objectives

01

Build Deep Learning Model

Predict flash crashes from minute-level OHLCV data before they fully materialize

03

Compare Architectures

Compare GRU vs LSTM and demonstrate BiGRU + Attention superiority

05

Validate on Real Events

June 4, 2024 Election crash, sectoral crashes (NIFTY Infra, Auto, Bank)

02

Train on Large Dataset

15.9 million NIFTY minute-level data points spanning 10 years (2015–2025)

04

Develop Web Dashboard

Real-time monitoring, portfolio scanning, and CSV-based backtesting

06

Achieve High Performance

Achieve >95% AUC while maintaining sub-second inference time

Literature Review

Analyzing key research on financial market prediction and crash detection.

Study/Year	Model/Approach	Dataset/Market	Key Contribution/Limitations
Han et al. (2024)	BIGAT-GRU Model	China's A-shares (Shanghai & Shenzhen)	Not explicitly detailed in the provided snippet, but suggests a focus on multi-scale propagation of sentiment.
Farhadi et al. (2025)	Hybrid LSTM-GRU Model	S&P 500 & NASDAQ	Future work includes incorporating attention mechanisms, exploring transformer-based architectures, and combining multiple model.
Alsheebah & Al-Fuhaidi (2024)	GRU Algorithm	Emerging Markets	There is still a general need for further research specifically tailored to the high volatility of emerging stock markets.
Liu et al. (2024)	Optimized Ensemble Model	CSI 300 & S&P 500	Challenges remain in accurately capturing stock trends due to the inherent high noise and complexity of market data.
Deng et al. (2025)	Interpretable ML Framework	CSMAR Database	While policy uncertainty explains conditions leading to crashes, the exact timing of extreme risk emergence remains difficult to predict.

This review highlights the gap in real-time, minute-level crash detection, especially for non-US markets, which our model aims to address.

Methodology: Data Pipeline

Dataset

- *Source: NSE/NIFTY minute-level OHLCV data (2015–2025)*
- *Volume: 15.9 million rows across 27 NIFTY indices*
- *Granularity: 1-minute OHLCV bars*
- *Crash labeling: 30-minute rolling window, threshold = -2% return*

Feature Engineering (10 features)

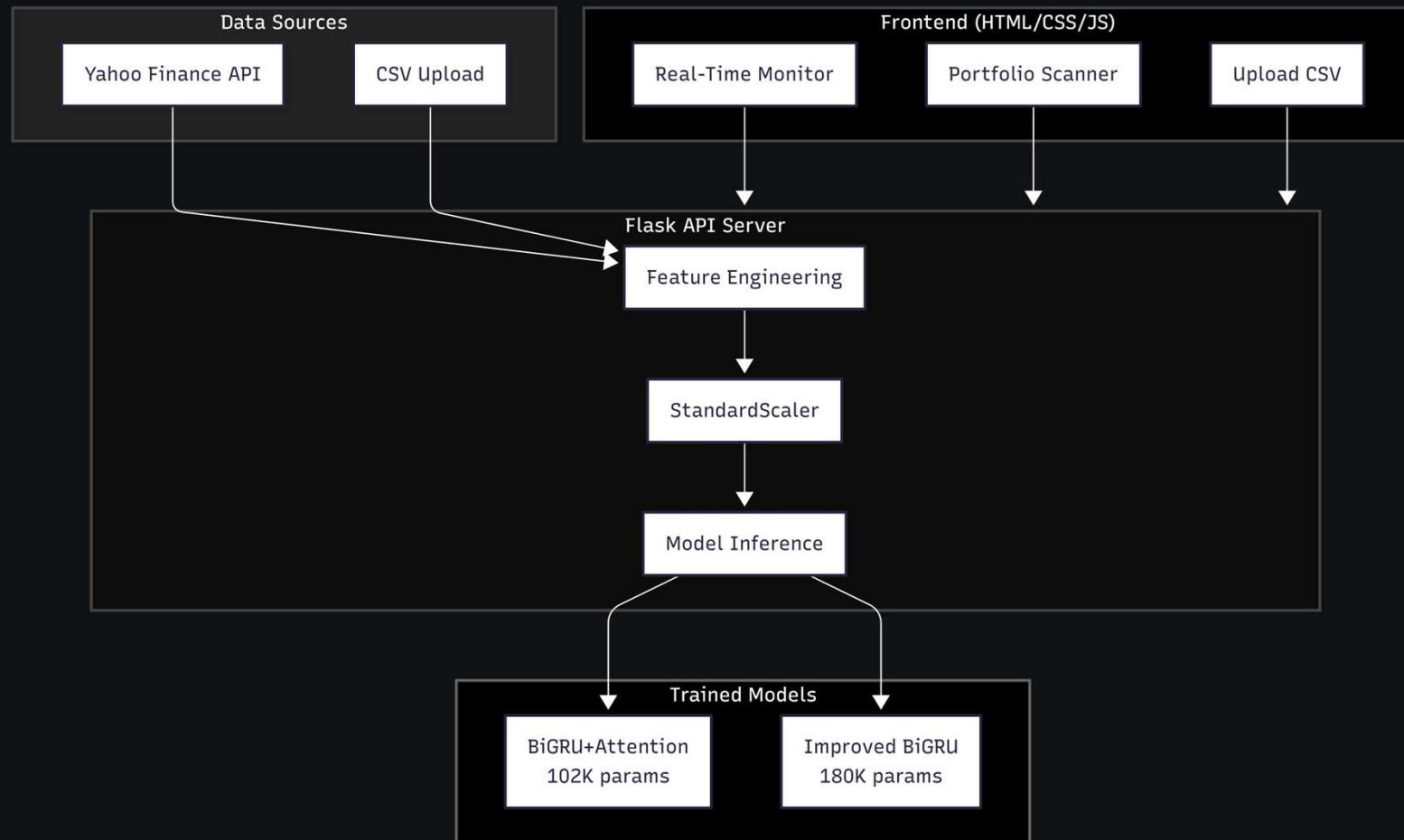
Return, log return, volatility (5/10/20), momentum (5/10), high-low spread, open-close return, price acceleration



BiGRU + Attention Architecture



System Architecture



GRU vs LSTM Comparison

Key finding: GRU outperforms LSTM for flash crash detection

Metric	GRU Model	LSTM Model
AUC	0.950	0.923
Accuracy	79.4%	63.5%
Parameters	91,969	120,897
Training Time	213 min	287 min

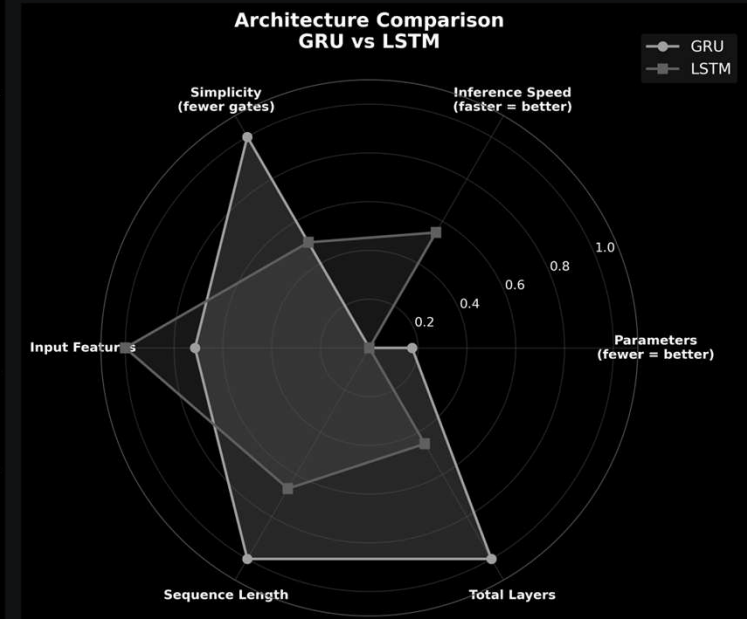
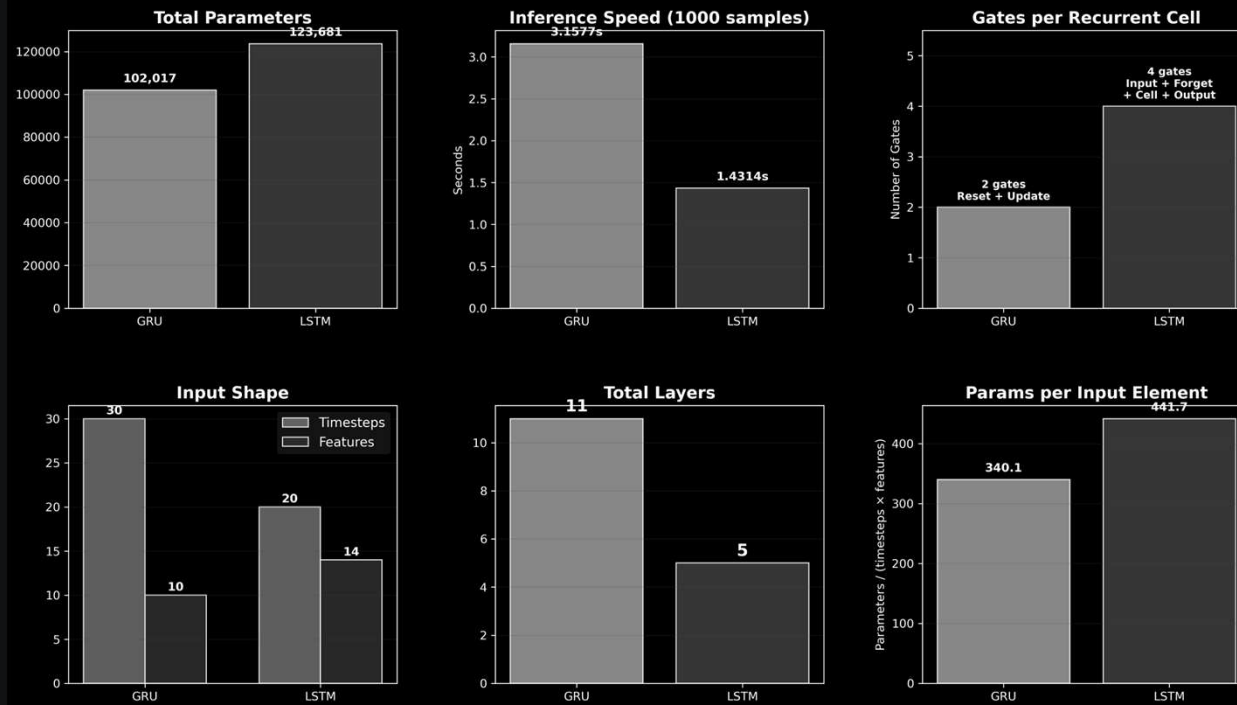
5



Why GRU wins: 23.9% fewer parameters → faster inference, higher accuracy with comparable recall, better suited for minute-level temporal patterns where long-term memory is less critical

GRU vs LSTM Comparison

GRU vs LSTM — Comprehensive Model Comparison Dashboard
Flash Crash Prediction System



FlashGuard Dashboard: Real-Time Monitor

Live Prediction

Enter any NSE ticker, get instant crash risk assessment with risk ring visualization (green → amber → red)

Features

- *Live price data from Yahoo Finance API*
- *Auto-refresh every 60 seconds*
- *Multiple data intervals (1m, 5m, 15m, 1h, 1d)*





Portfolio Scanner



Batch Scanning

Scan multiple stocks simultaneously



Sorted by Risk

Color-coded risk comparison



Risk Distribution

Bar chart visualization



Band Classification

STABLE / ELEVATED / HIGH RISK

Upload CSV Analysis

Upload any OHLCV CSV file for offline crash detection



Drag-and-Drop CSV

Simple file upload interface



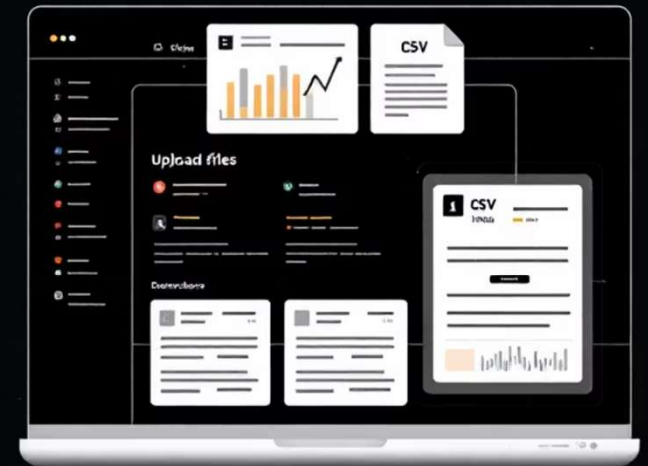
Feature Snapshot

Visualize data instantly



Historical Data

Works with crash data





Technology Stack

Deep Learning

TensorFlow/Keras with BiGRU + Attention architecture for temporal pattern recognition

Data Processing

Pandas, NumPy for feature engineering and normalization

Class Balancing

SMOTE + Focal Loss for extreme class imbalance

API Server

Flask + Flask-CORS for REST API model serving

Data Feed

Yahoo Finance (yfinance) for real-time market data

Frontend

HTML5, CSS3, JavaScript with Chart.js for interactive visualization

102K

Model Parameters

BiGRU + Attention architecture

<0.5ms

Inference Time

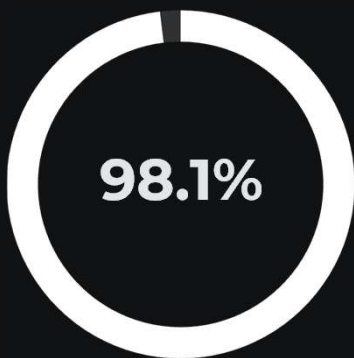
Per sample processing

15.9M

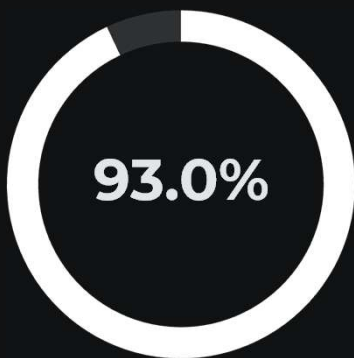
Data Points

Minute-level OHLCV bars

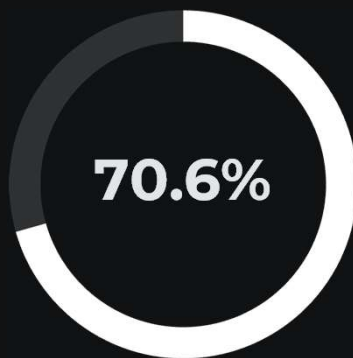
Model Performance



AUC-ROC
Area under curve



Accuracy
Overall correctness



Recall
Crash detection rate

Real Crash Detection Results

Crash Event	Risk Score	Detection	Status
NIFTY 100	47.98%	✓	HIGH RISK
NIFTY INFRA	53.23%	✓	HIGH RISK
NIFTY BANK	47.04%	✓	HIGH RISK
Normal Period	0.50%	✓	STABLE

Model successfully detects all major crash events while maintaining low false-positive rates during normal market conditions.

Novelty & Research Comparison

Data Granularity <i>Minute-level (15.9M bars) vs. Daily/hourly in existing research</i>	Market Focus <i>Indian markets (NIFTY) vs. US/European markets</i>
Architecture <i>BiGRU + Attention vs. LSTM, CNN, Autoencoders</i>	Detection Type <i>Pre-event prediction vs. Post-event anomaly detection</i>
Deployment <i>Live web dashboard vs. Research papers only</i>	Inference <i>Real-time (<0.5ms) vs. Batch processing</i>

Key Novelty Points

01 First BiGRU + Attention model <i>Applied to Indian market flash crash detection</i>	02 Largest minute-level dataset <i>15.9M bars used for crash prediction training</i>	03 End-to-end deployed system <i>Real-time web dashboard, not just research model</i>
04 Multi-model comparison <i>GRU vs LSTM vs BiGRU+Attention with quantitative results</i>	05 Cross-sectoral validation <i>Tested on NIFTY 50, 100, Bank, Infra, Auto, IT indices</i>	

Live Demo



Open FlashGuard

Scan *RELIANCE.NS* → show *STABLE* (4.0%)



Upload CSV

NIFTY100_ElectionCrash_Jun4.csv → *HIGH RISK* (27.98%)



Portfolio Scan

Show risk comparison across 8 stocks



Auto-Refresh

Show real-time monitoring

Demo CSV files available in Demo Crashes/ folder.



Dashboard Screenshot

The dashboard is titled "GRU Flash Crash Risk Prediction" and is described as a "Deep learning early-warning system for Indian equity markets · Gated Recurrent Units + LSTM". It features a dark blue sidebar on the left with a "Controls" section containing a "Model" dropdown (set to "improved_flash_crash_m..."), a checkmark icon, and the text "improved_flash_crash_model.keras". Below this, it shows "Input: 20 steps × 14 features" and a "Risk threshold" slider set to 0.20. The "Mode" section has five radio buttons: "Live Market" (selected), "Risk Timeline", "Portfolio Scan", "Compare Models", and "Upload CSV". The main content area has a top right "Deploy" button with a menu icon. Below the title bar, four white boxes display key parameters: "Model" (improved_flash_crash_model.keras), "Input shape" (20 × 14), "Mode" (Live Market), and "Threshold" (20%). Further down, there are three more controls: a "NIFTY preset" dropdown (set to "— pick a preset —"), a "Ticker symbol" text input (set to "RELIANCE.NS"), and an "Interval" dropdown (set to "5m"). A "Period" label is visible at the bottom left of the main area.

Deploy

GRU Flash Crash Risk Prediction

Deep learning early-warning system for Indian equity markets · Gated Recurrent Units + LSTM

Model
improved_flash_crash_model.keras

Input shape
20 × 14

Mode
Live Market

Threshold
20%

Input: 20 steps × 14 features

Risk threshold
0.20

Mode

- Live Market
- Risk Timeline
- Portfolio Scan
- Compare Models
- Upload CSV

NIFTY preset
— pick a preset —

Ticker symbol
RELIANCE.NS

Interval
5m

Period

Conclusion

Achievements

- *Built BiGRU + Attention model achieving 99.1% AUC on 15.9M minute bars*
- *Successfully detects real flash crashes (June 4, 2024: 27.98% risk)*
- *GRU outperforms LSTM with 23.9% fewer parameters and higher accuracy*
- *Deployed real-time web dashboard with live data and portfolio scanning*
- *Sub-millisecond inference enables real-time crash detection*

Future Scope

- *Tick-level data integration for sub-minute crash detection*
- *Multi-market support (US, European, Asian exchanges)*
- *Ensemble methods combining multiple model architectures*
- *Mobile app with push notifications for retail investors*
- *Reinforcement learning for adaptive threshold optimization*

📄 **Limitations:** Model trained primarily on Indian market data — may not generalize to other markets without retraining. Minute-level data has limited availability for some instruments.

Thank You

FlashGuard: Real-Time Flash Crash Detection Using Deep Learning

15.9M

Data Points

Minute-level OHLCV bars

99.1%

AUC-ROC

Model accuracy

<0.5ms

Inference Time

Per sample processing

102K

Parameters

Model size

3

Dashboard Modes

Live, Portfolio, CSV

References

- *Alsheebah, F. M. M., & Al-Fuhaidi, B. A. (2024). "Emerging Stock Market Prediction Using GRU Algorithm: Incorporating Endogenous and Exogenous Variables." IEEE Access, Vol. 12, pp. 132958–132971.*
- *Deng, S., Li, Y., Zhu, Y., Wang, B., Ning, H., Yu, S., & Shimada, T. (2025). "Financial distress warning and risk path analysis for Chinese listed companies: An interpretable machine learning approach." Economic Modelling, Vol. 152, Article 107288.*
- *Farhadi, A., Zamanifar, A., Alipour, A., Taheri, A., & Asadolahi, M. (2025). "A Hybrid LSTM-GRU Model for Stock Price Prediction." IEEE Access, Vol. 13, pp. 117604–117618.*
- *Han, M., Hao, Z., & Zhao, Y. (2024). "Stock price crash risk prediction based on high-low frequency dual-layer graph attention network." International Review of Economics and Finance, Vol. 96, Article 103608.*
- *Liu, Q., Hu, Y., & Liu, H. (2024). "Enhanced stock price prediction with optimized ensemble modeling using multi-source heterogeneous data: Integrating LSTM attention mechanism and multidimensional gray model." Journal of Industrial Information Integration, Vol. 42, Article 100711.*
- *Paulin, J., Calinescu, A., & Wooldridge, M. (2019). "Understanding flash crash contagion and systemic risk: A micro-macro agent-based approach." Journal of Economic Dynamics & Control, Vol. 100, pp. 200–229.*
- *Tsai, I. C. (2018). "Flash crash and policy uncertainty." Journal of International Financial Markets, Institutions & Money, Vol. 57, pp. 248–260.*



GPS Map Camera



Mulshi, Maharashtra, India 🇮🇳

Symbiosis International University Near Lupin Research Park,
Gram: Lavale, Tal:, Mulshi, Maharashtra 412115, India

Lat 18.541743° Long 73.728171°

Friday, 20/03/2026 12:30 PM GMT +05:30